

JOONWON LEE

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August 7, 2026

Goldman Sachs, Corporate Planning & Management
New York, NY

Re: Associate, Quantitative Strategist, Core Planning and Analysis Strats

Dear Core Planning and Analysis Strats Hiring Team,

I am applying for the Associate Quantitative Strategist position within the Core Planning and Analysis Strats team. I am a Statistics Ph.D. candidate at Rutgers University, expecting to graduate in January 2027. My background combines stochastic and time-dependent modeling, simulation, statistical diagnostics, model validation, and scalable Python engineering.

At JPMorgan Chase, I developed and validated a Sequential Probability Ratio Test for binary risk indicators, translating a monitoring requirement into explicit hypotheses and statistically controlled decision rules. I used Brownian-motion approximations, dynamic-programming probability propagation, and simulation to evaluate stopping behavior, Type I/II error, and model limitations. At Travelers, I built an end-to-end LightGBM pricing pipeline across more than 2.46 million records, benchmarked it against a generalized linear model, and communicated the results to business stakeholders.

My doctoral research develops scalable likelihood-based inference and diagnostics for large, dependent datasets. I built an advection-aware Vecchia approximation that fits up to 145,008 observations per day, designed simulation-based diagnostics that identify model failures by frequency scale, and implemented restartable Python/PyTorch workflows on CPU, GPU, and HPC systems. I also integrated compiled C++ ordering routines into Python through pybind11. Separately, I studied the effect of the Earned Income Tax Credit on household labor supply using fixed-effects regression and propensity-score matching.

I am drawn to this role's combination of quantitative planning models, rigorous validation, and analytical automation. My experience building modular research pipelines provides a strong foundation for production modeling, and I would be excited to extend that foundation to agentic systems for automated analysis and reporting. Thank you for your consideration.

Sincerely,

Joonwon Lee